

Group-Fair Stochastic Gradient Descent via Masked Optimal Transport

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ABSTRACT

Minibatch SGD is group-blind. Which groups appear in a batch is decided by how often they occur in the data, not by how much they are meant to count in the objective, and when the two disagree the iterates converge to a prevalence-weighted surrogate rather than to the target risk. The gap is a bias, so no amount of training removes it. A minibatch is a partial-participation event over groups, which makes group-fair training an instance of federated aggregation under availability skew, and we carry the masked optimal transport construction of FedAVOT across. Aligning the batch-composition distribution q with the target importance distribution p produces per-batch group weights whose induced stochastic sub-gradient is exactly unbiased for the target risk, giving the standard $O(1/\sqrt{T})$ rate in the nonsmooth convex regime with a constant that does not depend on how many groups a batch happens to cover. Exact alignment is feasible precisely under a Hall-type condition, which reduces to the per-group requirement that importance not exceed observation rate; past it Sinkhorn returns an information projection and a residual we compute in closed form rather than bound. That closed form turns the excess risk into two terms, an alignment gain fixed by the instance and an optimization residual paid by the estimator, and a severity sweep over eleven availability skews on Adult and IMDb-Wiki shows the two crossing: transport wins while the gain exceeds the residual and loses past it, at a threshold the decomposition locates in advance. The residual is a stepsize artifact, not a property of the method, and a decaying schedule removes it on every instance where it had reversed the sign. A mean-CVaR layer composed on top reaches the worst group at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$.

Index Terms— Group fairness, optimal transport, stochastic gradient descent, distribution shift, conditional value at risk.

1 INTRODUCTION

A classifier is rarely asked to minimize one risk. It is asked to minimize one while behaving acceptably on sub-populations the training distribution under-represents: a diagnostic model across patient groups, a recognition system across demographic strata [1, 2, 3]. The standard response is to change the objective, replacing the empirical risk by $F_p(\theta) = \sum_i p_i f_i(\theta)$ with p the weighting one actually wants, uniform, inverse to prevalence, or externally dictated [4, 5].

This paper is about a gap between writing that objective down and optimizing it. Minibatch SGD does not see p : it sees a batch, whose group composition is governed by prevalence and by the sampler, so averaging the per-example gradients descends a different function. The estimator is not

merely noisy, it is biased, by a fixed vector that survives $T \rightarrow \infty$ and any stepsize schedule. Rebalancing the sampler is the usual and only partial fix: a group whose prevalence is below the resolution of the batch size cannot be sampled into every batch however the sampler is tilted, and upscaling the few examples that do appear reintroduces an amplitude distortion that destabilizes the iterates [6].

Groups as clients. The same mismatch is the central object of a different literature. In federated learning the *availability distribution* q over which clients participate in a round rarely matches the *importance distribution* p that defines the objective, and FedAvg consequently optimizes a surrogate whose weights are induced by the availability process rather than chosen [7, 8, 9]. FedAVOT [6] removes that mismatch by posing aggregation as a masked optimal transport (MOT) problem: find a coupling whose column marginal is the availability law and whose row marginal is the target p , supported only on client-round pairs that actually occurred. The observation driving this paper is that a minibatch is a participation event: the groups represented in B^t are the active set S^t , its law is q , and the weighting one wants is p . Group-fair SGD is federated aggregation with the federation collapsed onto one machine. We call the method GAVOT and keep the FedAVOT label in figures, the aggregation rule being unchanged.

Contributions.

1. *The reduction* (Sec. 2). Group-blind minibatch SGD reduces, under uniform-within-batch averaging, to agnostic FedAvg with one local step, so the surrogate identity of [7] names the function SGD actually minimizes, and transport weights make the batch subgradient exactly unbiased for ∇F_p (Lemma 5).
2. *A coverage law and a severity measure* (Cor. 4). Feasibility is Hall-type and reduces on singletons to $p_i \leq r_i$, importance below observation rate, so the *infeasible mass* $\nu \triangleq \sum_{i:p_i > r_i} p_i$ is the natural severity axis, computable before training.
3. *A two-term decomposition that locates the crossover* (Thm. 8, Sec. 5). Past the Hall condition the excess splits into an *alignment gain* Δ_{floor} , the distance between the two optima each rule actually converges to, and an optimization residual. The gain is exact and instance-fixed; the residual is paid by the estimator. We sweep eleven skews and watch the two cross.
4. *Risk-averse layer* (Sec. 4). Mean-CVaR over groups composes with the transport weights without disturbing unbiasedness, at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$, $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$.

Related work. Reduction methods for group fairness return randomized mixtures or carry an approximation constant tied to a fixed class [11, 12, 13]. Distributionally robust

and worst-group objectives [3, 2, 28] change the functional rather than the sampler and are complementary: our CVaR layer is one such functional, obtained on top of transport rather than instead of it. Importance sampling for SGD [16, 17] corrects variance, not the identity of the minimized objective.

2 PROBLEM FORMULATION

Groups and objectives. Let the population be partitioned into N groups indexed by $i \in [N]$ with *prevalences* $\pi \in \Delta^{N-1}$, $\pi_i = \mathbb{P}[\text{group } i]$. Group i carries a distribution $\mathcal{D}_i$ on $\mathcal{X} \times \mathcal{Y}$ and the risk $f_i(\theta) \triangleq \mathbb{E}_{(X,Y) \sim \mathcal{D}_i}[\ell(m(X; \theta), Y)]$, $\theta \in \mathcal{C} \subseteq \mathbb{R}^{d'}$, with $m(\cdot; \theta)$ the predictor and ℓ the loss. Empirical risk minimization weights the groups by prevalence, $F_\pi = \sum_i \pi_i f_i$, so a majority group dominates the trained model. The remedy is to fix an *importance distribution* $p \in \Delta^{N-1}$ and minimize

$$F_p(\theta) \triangleq \sum_{i=1}^N p_i f_i(\theta), \quad \theta^* \in \arg \min_{\theta \in \mathcal{C}} F_p(\theta), \quad (1)$$

uniform p being the category-uniform objective, $p_i \propto 1/\pi_i$ the inverse-prevalence one, or p a fairness constraint [4, 5].

Assumption 1. $\mathcal{C}$ is convex and compact with diameter D ; each f_i is convex on $\mathcal{C}$ and $0 \leq f_i \leq M$; every stochastic subgradient $g_i(\theta)$ of f_i satisfies $\mathbb{E}\|g_i(\theta)\|^2 \leq G^2$ on $\mathcal{C}$.

These are the assumptions of [7, 6] restated for groups; they restrict neither p nor π .

Prevalence versus importance. Minibatch SGD does not see p . At step t the sampler draws a batch B^t of b examples; the *active set* $A^t \triangleq \{i : B^t \text{ contains an example from group } i\}$ has support $\{A_j\}_{j \in [M]}$ and *composition law* $q_j \triangleq \mathbb{P}[A^t = A_j]$, known or estimated offline by simulating the sampler. The observation rate of group i is $r_i \triangleq \mathbb{P}[i \in A^t] = \sum_{j: i \in A_j} q_j$, under i.i.d. sampling $1 - (1 - \pi_i)^b$: a group whose prevalence is below the resolution of the batch is absent from most batches whatever p says. This is federated learning under partial participation, with A^t the participating clients, q the availability and p the importance distribution [6, 7].

What group-blind SGD minimizes. The group-blind update averages the per-example subgradients of B^t , giving group $i \in A^t$ the weight $\hat{\pi}_i^t$, its share of the batch, and absent groups zero. Averaging over the batch given A^t and then over $A^t \sim q$, the hierarchical experiment of [7, Eq. 4], names the function SGD in fact minimizes.

Proposition 2 (Surrogate objective). *Under Assumption 1, group-blind minibatch SGD is a stochastic subgradient method on $F_{\tilde{p}}$ with*

$$\tilde{p}_i = \sum_{j: i \in A_j} q_j \mathbb{E}[\hat{\pi}_i^t | A^t = A_j], \quad (2)$$

a probability vector. For i.i.d. sampling $\tilde{p} = \pi$; for uniform weighting within the batch $\tilde{p}_i = \sum_{j: i \in A_j} q_j / |A_j|$, the surrogate of agnostic FedAvg [6, Eq. 2]. Consequently, for every $\theta \in \mathcal{C}$,

$$|F_p(\theta) - F_{\tilde{p}}(\theta)| \leq M \|p - \tilde{p}\|_1, \quad (3)$$

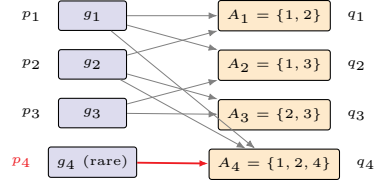


Figure 1: The mask E . Batch compositions carry the mass q they occur with; groups must receive the mass p the objective assigns them. Group 4 is reachable only through A_4 , so exact alignment requires $p_4 \leq q_4$: the Hall condition of Thm. 3 read on the singleton $I = \{4\}$. Shrinking the batch deletes A_4 from the support and makes p_4 unreachable.

and any method that minimizes $F_{\tilde{p}}$ exactly incurs the excess $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq 2M \|p - \tilde{p}\|_1$ on (1), which does not vanish with T .

(Proof in the appendix.) The choice of p in (1) is thus decorative unless the aggregation step changes: the bias $p - \tilde{p}$ survives every stepsize schedule, rebalancing the sampler cannot help a group with $p_i > r_i$ since $\tilde{p}_i \leq r_i$ always, and upscaling the examples that do appear destabilizes the iterates [6]. The question of [6], read on batches: *can the per-batch group weights be chosen so that SGD driven by q converges for the objective defined by p ?*

3 PROPOSED APPROACH: ALIGNMENT BY MASKED OPTIMAL TRANSPORT

Weights as a coupling. Let $Y[i, j] \geq 0$ be the weight of group i when the active set is A_j , $Y[i, j] = 0$ for $i \notin A_j$, and $T[i, j] \triangleq q_j Y[i, j]$ the joint mass carried from composition j to group i . Requiring the weights to be a probability vector on each realized composition and to reproduce p in expectation is the *masked optimal transport* (MOT) feasibility problem of [6],

$$T \mathbf{1}_M = p, \quad \mathbf{1}_N^\top T = q, \quad T \geq 0, \quad \text{supp}(T) \subseteq E, \quad (4)$$

with mask $E = \{(i, j) : i \in A_j\}$: a group absent from the batch cannot be weighted. Unlike classical transport [25, 20] there is no cost to minimize; any coupling respecting (4) serves. Fig. 1 is the picture.

When alignment is possible. Existence of a coupling is a supply-demand question on the bipartite graph E , settled by flow duality; proofs are in the appendix.

Theorem 3 (Feasibility, [6, Thm. 1]). *For $I \subseteq [N]$ let $\mathcal{N}(I) = \{j : \exists i \in I, (i, j) \in E\}$ and $\mathcal{S}(I) = \{j : A_j \subseteq I\}$. Problem (4) is feasible if and only if*

$$\sum_{j \in \mathcal{S}(I)} q_j \leq \sum_{i \in I} p_i \leq \sum_{j \in \mathcal{N}(I)} q_j \quad \forall I \subseteq [N]. \quad (5)$$

Corollary 4 (Coverage law and severity). *Feasibility of (4) requires $\mathbb{P}[A^t = \{i\}] \leq p_i \leq r_i$ for every i , with $r_i = \mathbb{P}[i \in A^t]$ the observation rate of Sec. 2. Writing*

$$\nu \triangleq \sum_{i: p_i > r_i} p_i \quad (6)$$

for the infeasible mass, $\nu = 0$ iff the upper constraints hold. Under i.i.d. sampling of b examples the upper constraint

Algorithm 1 GAVOT: group-fair SGD by masked transport

Require: target p , composition law $(q, \{A_j\}, E)$, step η , horizon T , CVaR parameters (α, γ) with $\gamma = 1$ for the risk-neutral method

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1:  $(T, Y) \leftarrow$  Sinkhorn scaling for  $(p, q, E)$   $\triangleright$  offline, once
2:  $\theta^0 \in \mathcal{C}$ ,  $t^0 \in \mathbb{R}$ 
3: for  $t = 1, \dots, T$  do
4:   draw batch  $B^t$ ; set  $A^t$ , let  $j(t)$  be s.t.  $A_{j(t)} = A^t$ 
5:    $g_i^t \leftarrow$  stochastic subgradient of  $f_i$  from  $B^t \cap$  group  $i$ 
6:    $w_i^t \leftarrow Y[i, j(t)](\gamma + \frac{1-\gamma}{\alpha} \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$ 
7:    $\theta^t \leftarrow \Pi_{\mathcal{C}}(\theta^{t-1} - \eta \sum_{i \in A^t} w_i^t g_i^t)$ 
8:    $t^t \leftarrow t^{t-1} - \eta(1 - \gamma)(1 - \frac{1}{\alpha} \sum_{i \in A^t} Y[i, j(t)] \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$ 
9: end for
Ensure:  $\theta_T = \frac{1}{T} \sum_{t=1}^T \theta^t$ 

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reads $b \geq \log(1 - p_i)/\log(1 - \pi_i)$; under K -of- N uniform group sampling it reads $p_i \leq K/N$. The infeasible mass is the axis Sec. 5 sweeps; feasible and infeasible regimes are its two ends.

Computing the plan. Given feasibility, a coupling is obtained by iterative proportional fitting (Sinkhorn scaling) on the masked support [18, 19, 20], rescaling rows to p and columns to q in turn (appendix, (11)), at $O(|E|)$ per sweep with $|E| \leq NM$ and M the number of *realized* compositions, once, offline; training then adds one lookup and one multiply per group per step. Since $\mathbf{1}_N^\top T = q$ and $Y = T \text{diag}(q)^{-1}$, every column $Y[\cdot, j]$ is a probability vector on A_j , the property each bound uses. Algorithm 1 is the resulting method; $\gamma = 1$ is the risk-neutral rule analyzed here, $\gamma < 1$ the layer of Sec. 4.

Analysis. Let $\hat{g}(\theta) = \sum_{i \in A^t} Y[i, j(t)] g_i(\theta)$ be the transport-weighted batch subgradient of Algorithm 1, g_i the stochastic subgradient of f_i from $B^t \cap$ group i . Everything rests on [6, Lem. 1] moved from models to gradients: transport removes bias; it is not a variance reduction device.

Lemma 5 (Unbiasedness). *Let T satisfy (4). Then $\mathbb{E}[\hat{g}(\theta) | \theta] \in \partial F_p(\theta)$ for every $\theta \in \mathcal{C}$.*

Lemma 6 (Second moment, free of coverage). $\mathbb{E}\|\hat{g}(\theta)\|^2 \leq G^2$ for every $\theta \in \mathcal{C}$, with no dependence on $|A^t|$ or on N .

Theorem 7 (Rate). *Under Assumption 1, if (4) is feasible then Algorithm 1 with $\gamma = 1$ and $\eta = D/(G\sqrt{T})$ satisfies*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) \leq \frac{DG}{\sqrt{T}}. \quad (7)$$

The constant is that of full-participation SGD, independent of how many groups a batch covers, though the realized variance is larger.

Outside the feasible region. When (5) fails, Sinkhorn does not diverge: its iterates accumulate at the coupling whose row marginal $\hat{p} \triangleq T\mathbf{1}$ is the KL-closest to p over the reachable polytope $\{T\mathbf{1} : T \geq 0, \mathbf{1}^\top T = q, \text{supp}(T) \subseteq E\}$ [22, 21, 20]. The column marginal is met exactly, so both lemmas hold with $\hat{p}$ for p : the algorithm is unbiased SGD on the surrogate $F_{\hat{p}}$, and on singletons $\hat{p}_i \leq r_i$, so groups with $p_i > r_i$ are truncated at their observation rate.

Theorem 8 (Infeasible regime). *Let $\hat{p}$ be the row marginal of the Sinkhorn limit, $\theta^\dagger \in \arg \min_{\mathcal{C}} F_{\hat{p}}$, and write $\Phi(w) \triangleq F_p(\arg \min_{\mathcal{C}} F_w)$ for the p -risk of the minimizer of the w -weighted risk. Then*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) = \underbrace{\Phi(\hat{p}) - \Phi(p)}_{\text{alignment floor}} + \underbrace{\varepsilon_T(\hat{p})}_{\text{optimization residual}}, \quad (8)$$

where $\varepsilon_T(\hat{p}) \triangleq \mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^\dagger)$ satisfies $\mathbb{E}F_{\hat{p}}(\bar{\theta}_T) - F_{\hat{p}}(\theta^\dagger) \leq DG/\sqrt{T}$ and $\varepsilon_T(\hat{p}) \rightarrow 0$ whenever $\bar{\theta}_T \rightarrow \theta^\dagger$. Group-blind averaging obeys (8) with $\tilde{p}$ in place of $\hat{p}$. Writing $\Delta_{\text{floor}} \triangleq \Phi(\tilde{p}) - \Phi(\hat{p})$ for the alignment gain, transport improves on group-blind averaging exactly when $\Delta_{\text{floor}} > \varepsilon_T(\hat{p}) - \varepsilon_T(\tilde{p})$, and every floor is bounded by $\Phi(w) - \Phi(p) \leq 2M\|p - w\|_1$.

Both terms are available before training, $\hat{p}$ and $\tilde{p}$ from (p, q, E) and Φ exactly on instances with a closed-form or convex frontier; the gain decides the sign, the residual is a stepsize artifact (Sec. 5), and the ℓ_1 bound is too loose to discriminate.

4 A RISK-AVERSE LAYER OVER GROUPS

Exact alignment delivers the p -weighted risk, an average, which can hide a group. Where the requirement is on the worst group instead, we compose a mean-CVaR functional over groups on top of the transport weights [7, 26]:

$$F_p^{\alpha, \gamma}(\theta) \triangleq \gamma F_p(\theta) + (1 - \gamma) \text{CVaR}_\alpha^p(f(\theta)), \quad \alpha, \gamma \in (0, 1], \quad (9)$$

where CVaR_α^p is the conditional value at risk of the group-loss vector under p . The Rockafellar–Uryasev representation turns (9) into a joint minimization over (θ, t) ,

$$\Phi(\theta, t) = \gamma \sum_i p_i f_i(\theta) + (1 - \gamma) \left(t + \frac{1}{\alpha} \sum_i p_i [f_i(\theta) - t]_+ \right), \quad (10)$$

jointly convex, with subgradients in both arguments again p -weighted sums of per-group quantities, so Lemma 5 applies unchanged. This is why the layers compose rather than interfere: transport fixes *which* distribution the expectation is under, CVaR changes *what* is averaged. Lines 6 and 8 of Algorithm 1 implement (10).

Theorem 9 (Rate under mean-CVaR). *Under the hypotheses of Theorem 7, Algorithm 1 with $\eta = \Theta(1/\sqrt{T})$ satisfies $\mathbb{E}[F_p^{\alpha, \gamma}(\bar{\theta}_T)] - \min F_p^{\alpha, \gamma} = O(\kappa(\alpha, \gamma)/\sqrt{T})$ with $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$. In the infeasible regime the bound of Theorem 8 acquires the same factor on its statistical term, the bias term being unchanged.*

Both $\gamma = 1$ and $\alpha = 1$ recover Theorem 7 exactly, so the grid contains the risk-neutral method along two edges, visible as the constant row and column of every heatmap below. Sending $\alpha \downarrow 0$ drives (9) to the worst group at the cost $\kappa(\alpha, \gamma) \rightarrow \infty$: the parameter prices robustness in the convergence constant, not in the limit.

Remark 10 (Relation to constrained learning). $\max_i f_i$ is the exact-penalty limit of $\min f_0$ s.t. $f_i \leq c_i$, a fixed price on the largest violation matching penalized and constrained values once it exceeds the least ℓ_1 norm on the dual-optimal set [15]. The $\alpha \downarrow 0$ edge of (9) is thus the scalarization a constrained specification selects, with κ its price in the rate.

Table 1: Headline cells, tail-500 over 5 seeds, decaying step-size. *full* sees every group every step. The fixed multiplier m/K diverges whenever observation rates are non-uniform.

Instance		GAVOT	unif. avg	m/K	full
Adult, prevalence ($\nu=.60$)	0.2269	0.2479	0.670	0.2073	
Adult, aligned ($\nu=0$)	0.2075	0.2077	div.	0.2073	
IMDb, skewed ($\nu=.31$)	86.06	91.63	8504	82.95	
IMDb, aligned ($\nu=0$)	84.28	86.37	div.	82.95	

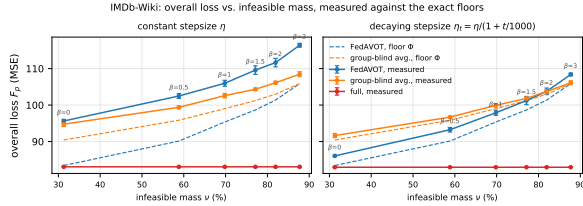


Figure 2: IMDb-Wiki severity sweep. Solid, measured tail-500 (± 1 std); dashed, the exact floor $\Phi(\cdot)$ each rule converges to. Left, constant stepsize: the optimization residual swamps the alignment gain and transport losses everywhere, even where the floors are seven units apart. Right, decaying stepsize: the residual collapses and the measured curves track their floors, so transport wins until the floors close at high ν . The sign is set by which of the two terms of (8) dominates, not by the regime label.

5 EXPERIMENTS

Instances and protocol. *Adult* income classification [33]: logistic regression ($\eta=0.1$) on 100 race-homogeneous groups of 30 samples, groups per race proportional to prevalence (85/10/3/1/1), target uniform over the five races, so a group from the smallest race carries $p_i = 0.2$. *IMDb-Wiki* age regression [34]: linear regression on 128-d face embeddings ($\eta=10^{-2}$) over 100 identities with the cubic target $p_i \propto (101-i)^3$. Observation rates are swept: on Adult $r \propto p^\beta$, $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, from uniform observation (prevalence, $\nu=.60$) to aligned ($\nu=0$); on IMDb-Wiki $r_i \propto i^\beta$, $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, from uniform ($\nu=.31$) to the mirror image of p ($\nu=.88$), plus the aligned $r_i \propto 101-i$ ($\nu=0$): eleven instances. Each step observes $K=3$ groups drawn $\propto r$; the plan is fitted once per instance by Sinkhorn scaling on the mask. Each observed group takes $H=5$ local steps before the weighted average, the standard practical deviation from the one-step analysis, applied to every rule alike: **GAVOT**; *group-blind averaging*, the uniform average over the observed groups ($\tilde{p}_i = r_i/K$); the *fixed multiplier* m/K ; and *full*, every group every step weighted by p . Stepsize $\eta_t = \eta/(1+t/1000)$ unless stated, 4000 steps, 5 seeds, F_p over the last 500 steps, mean $\pm$ std. The floors of Theorem 8 are exact (closed form for least squares, Newton for the logistic loss). Full protocol and the closed-form check of the mirrored instance are in the appendix; the extended version carries the unabridged discussion.

E1: the headline cells. Table 1 reports the four instances a practitioner would recognize. Transport wins on three: Adult under prevalence-proportional availability,

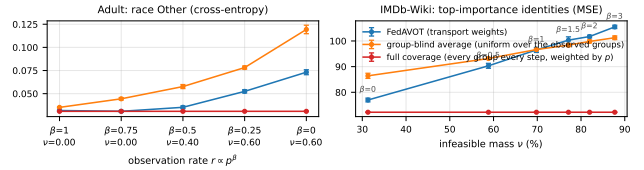


Figure 3: Loss on the least represented group, decaying step-size. Left, Adult: cross-entropy on race Other (1 group of 30, $p_i = 0.2$) against the observation-rate exponent. Right, IMDb-Wiki: MSE on the 20 highest-importance identities, the least observed once $\beta > 0$. Same runs as Table 1; constant-stepsize panels in the extended version.

Table 2: IMDb-Wiki severity sweep, decaying stepsize. Δ_{floor} is the alignment gain of Theorem 8, *gain* the measured advantage of transport, t its Welch statistic (population std). The excess residual transport pays averages 1.9; the sign flips where Δ_{floor} crosses it. Adult skews: appendix.

β	ν	Δ_{floor}	gain	t
0	.31	6.99	+5.57	20.7
0.5	.59	5.71	+3.46	11.3
1.0	.70	3.67	+2.05	5.4
1.5	.77	2.56	+0.58	1.1
2.0	.82	1.62	−0.03	−0.1
3.0	.88	0.02	−2.28	−7.9

0.2269 against 0.2479 ($t=45$), and both IMDb cells, 86.06 against 91.63 ($t=21$) and 84.28 against 86.37 ($t=13$). On Adult with aligned availability the two coincide at the oracle value, as Proposition 2 predicts when $\tilde{p} = p$. The fixed multiplier diverges on both aligned instances and is far off on the other two: naive rescaling is no alternative to solving (4).

E2: the least represented group. Fig. 3 reads the same runs on the group the target exists for. On Adult, transport lowers the loss on race Other at every skew and under both stepsizes: .073 against .119 on the prevalence instance (full .031), .031 against .035 when aligned, matching full; Amer-Indian behaves the same way. On IMDb-Wiki the top-importance identities show the crossover of Table 2 one step earlier: under decay transport wins at $\nu \leq .70$ (77.0 against 86.4 at $\nu=.31$, full 72.2; 73.6 against 79.9 aligned) and loses at the three most infeasible skews. The single most starved group (largest p_i/r_i) is Other itself on Adult ($p_i/r_i=6.7$) and the top identity on IMDb-Wiki: 253 against 331 at $\nu=.31$, 373 against 380 at $\nu=.59$ (full 219), a 30-image group so ± 5 to 10 across seeds. The rare-group loss tells the same story as the objective, more sensitively.

E3: the crossover, and where the sign comes from. Table 2 and Fig. 2 are the main evidence. On Adult the measured gain tracks Δ_{floor} within 4% at three of five skews (.0134 against .0129, .0050 against .0049, .0002 against .0003) and exceeds it at the two most infeasible (.0210 against .0110, .0181 against .0144). On IMDb-Wiki the gain sits below the floor by a residual close to constant in β , mean 1.9, and the sign flips exactly where Δ_{floor} crosses that value, between $\beta=1.5$ (2.56, gain +0.58) and $\beta=2.0$ (1.62, −0.03):

Theorem 8 at finite T . The ℓ_1 criterion cannot do this: $\|p - \hat{p}\|_1 < \|p - \tilde{p}\|_1$ holds in all eleven instances, including the three where transport loses or ties.

E4: why the sign flips. At constant stepsize (Fig. 2, left) transport loses on all six IMDb skews, by up to 7.9, even where the floors are 6.99 apart. A starved group has $p_i > r_i$, so the plan delivers at most r_i of its mass: $\hat{p}$ truncates at the observation rate and, as ν grows, $\hat{p}$ and $\tilde{p}$ starve the same groups, so the floors converge (Δ_{floor} from 6.99 to 0.02; both 105.9 at $\nu=.88$, appendix). What transport still does is concentrate weight on the starved group whenever it appears, p_i/r_i times its batch share: Lemma 6 bounds the second moment identically for both rules, but the variance about the mean is larger, largest on the rare group, and at constant stepsize the iterates sit on a noise floor proportional to it. Under $\eta_t = \eta/(1 + t/1000)$ that floor decays, the curves fall onto their dashed floors, and what remains is the residual, 2.3 at $\nu=.88$, the one significant loss. Transport helps exactly while there is mass left to move.

E5: risk aversion, and where it does not help. On a 10×10 (α, γ) grid the overall objective is minimized on the $\gamma = 1$ edge of every instance, $\kappa(1, \gamma) = 1$ made visible: risk aversion costs the average. It earns its keep once, on the worst race of Adult under prevalence availability: GAVOT+CVaR at $(0.2, 0.7)$ reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging. Per race, transport improves four of five (Black .179 vs .200, Asian-Pac .368 vs .399, Am-Ind .175 vs .189, Other .073 vs .119) and regresses only the majority (.339 vs .332). On the least represented group the layer never helps: the grid’s best rare-group loss is at $\gamma=1$ on all four headline instances, and $(0.1, 0.1)$ nearly triples the loss on Other (.207 against .073) and lifts the IMDb-Wiki top identities from 77.0 to 103. CVaR tilts among the groups present in the batch, and the rare group is absent from most, so the tilt lands elsewhere; on top of transport it multiplies weights already p_i/r_i large and amplifies variance, not alignment. Risk aversion changes the functional, not who is observed.

6 CONCLUSION

Group-fair training fails in a specific and fixable place: the aggregation step, blind to the weighting the objective declares. Reading a minibatch as a partial-participation event makes the fix the masked transport problem that corrects availability skew in federated learning, with the same feasibility characterization, an $O(1/\sqrt{T})$ guarantee free of batch coverage, and a coverage law whose infeasible mass ν orders instances by severity. Past feasibility the excess splits cleanly, into an alignment gain fixed by the instance and an optimization residual paid by the estimator, and a sweep over eleven skews shows the two crossing where the decomposition says they will. Transport is worth applying while the gain exceeds the residual, both are computable before training, and the residual is a stepsize artifact a decaying schedule removes. The nonconvex case remains open.

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A APPENDIX

A. Proofs. *Proposition 2.* Conditionally on $A^t = A_j$ the within-group averages are unbiased for ∂f_i , so $\mathbb{E}[\text{batch subgradient} \mid A^t = A_j] = \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] \partial f_i(\theta)$; averaging over q gives $\sum_i \tilde{p}_i \partial f_i = \partial F_{\tilde{p}}$, and $\sum_i \tilde{p}_i = \sum_j q_j \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t \mid A_j] = 1$. Both bounds follow from $0 \leq f_i \leq M$ and the two-point inequality $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq [F_p - F_{\tilde{p}}](\theta_{\tilde{p}}^*) + [F_{\tilde{p}} - F_p](\theta^*)$.

Sinkhorn scaling. Starting from any $T^{(0)} > 0$ on E with $\mathbf{1}^\top T^{(0)} = q$, alternate

$$T \leftarrow \text{diag}(p \oslash T \mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (11)$$

until $\|T \mathbf{1} - p\|_1 \leq \varepsilon$. Under (5) the iteration converges to a solution of (4) [18, 22], being block-coordinate ascent on the dual of the entropic problem $\min\{\sum_E T \log T : T \mathbf{1} = p, \mathbf{1}^\top T = q\}$.

Lemma 5. Conditionally on $A^t = A_j$, $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$ for $i \in A_j$. Averaging over $A^t \sim q$, $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$, the row-marginal constraint being used in the last step.

Lemma 6. $Y[\cdot, j]$ is a probability vector on A_j , so $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$ by convexity of $\|\cdot\|^2$; take expectations and use $\sum_j T[i, j] = p_i$: $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$.

Theorem 3 (sketch). Route supply p_i from a source through group i , along the edges of E with infinite capacity, into composition j with demand q_j , and to a sink. A coupling exists iff the maximum flow saturates every source edge, which by max-flow/min-cut [24] is the right inequality of (5); the left one is the same statement for the reversed network. On singletons (5) is Hall’s condition [23].

Corollary 4. For $I = \{i\}$, $\mathcal{N}(I)$ is the set of compositions containing i , of total mass r_i , and $\mathcal{S}(I) = \{j : A_j = \{i\}\}$. Substitute $r_i = 1 - (1 - \pi_i)^b$ and $r_i = K/N$.

Theorem 7. By non-expansiveness of Π_C , $\|\theta^t - \theta^*\|^2 \leq \|\theta^{t-1} - \theta^*\|^2 - 2\eta(\hat{g}^t, \theta^{t-1} - \theta^*) + \eta^2 \|\hat{g}^t\|^2$. Conditioning on θ^{t-1} , Lemma 5 and convexity bound the inner product below by $F_p(\theta^{t-1}) - F_p(\theta^*)$, and Lemma 6 bounds the last term by $\eta^2 G^2$. Summing, telescoping, and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, and the stated η balances the two terms [30]. In full: $\mathbb{E}\|\theta^t - \theta^*\|^2 \leq \mathbb{E}\|\theta^{t-1} - \theta^*\|^2 - 2\eta \mathbb{E}[F_p(\theta^{t-1}) - F_p(\theta^*)] + \eta^2 G^2$; summing over $t \leq T$, telescoping, dropping the terminal distance and applying Jensen to $\bar{\theta}_T$ gives $\mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^*) \leq D^2/(2\eta T) + \eta G^2/2$, which is $DG/\sqrt{T}$ at $\eta = D/(G\sqrt{T})$; the same bound holds for $\eta_t = \eta_0/\sqrt{t}$ up to constants.

Theorem 8. Add and subtract $F_p(\theta^\dagger) = \Phi(\tilde{p})$; the floor is exact by definition and $\Phi(p) = F_p(\theta^*)$. Theorem 7 applied to $F_{\tilde{p}}$ gives the residual bound on the surrogate; the residual on F_p differs from it by $(F_p - F_{\tilde{p}})(\bar{\theta}_T) - (F_p - F_{\tilde{p}})(\theta^\dagger)$, which vanishes as $\bar{\theta}_T \rightarrow \theta^\dagger$ (for instance under strong convexity, which both of our instances satisfy). The ℓ_1 bound is the two-point argument of Proposition 2 with w for $\tilde{p}$.

B. The infeasible regime in closed form (IMDb-Wiki, $\beta = 3$). On the mirrored instance, where 41 of 100 groups violate $p_i \leq r_i$ and hold 88% of the importance mass, the Sinkhorn limit truncates the target at the observation-rate ceiling, $\tilde{p}_i \approx \min(p_i, r_i)$ up to renormalization (maximum

deviation 0.023), giving $\|p - \tilde{p}\|_1 = 1.58$ of a maximum possible 2. Because the task is linear regression, the minimizers θ^* of F_p and $\theta^\dagger$ of $F_{\tilde{p}}$ are available in closed form: $\Phi(p) = F_p(\theta^*) = 82.9$ (matching the measured full-coverage plateau, 82.95) and $\Phi(\tilde{p}) = F_{\tilde{p}}(\theta^\dagger) = 105.9$. The alignment floor is therefore 23.0, and the realized bias equals its exact expression $|(p - \tilde{p})^\top L(\theta^\dagger)| = 30.7$, where $L(\theta)$ collects the per-group risks. The ℓ_1 form $2M\|p - \tilde{p}\|_1$ of Theorem 8 holds but is conservative here, since any uniform bound satisfies $M \geq \max_i L_i(\theta^\dagger) \approx 454$. The group-blind surrogate $\tilde{p}_i = r_i/K$ has the same floor to two decimals ($\Phi(\tilde{p}) = 105.9$); both surrogates starve the same groups, which is why Δ_{floor} vanishes at this endpoint of Table 2.

E. Instances and protocol, in full. Adult: logistic regression with $d=109$ features including an intercept, cross-entropy loss; 100 groups of 30 samples, each homogeneous in race, with 85/10/3/1/1 groups for White/Black/Asian-Pac./Amer-Ind./Other; the target gives each race 1/5, split equally among its groups, so on the prevalence instance the five groups of the three smallest races hold 60% of the target against $r_i \approx K/m = .03$. IMDb-Wiki: the first 100 identities with 30 images, 128-d ResNet embeddings, linear regression of age; at $\beta=0$ nine groups are infeasible (mass .31), at $\beta=3$ forty-one (.88). Each step draws $K=3$ groups without replacement with probabilities $\propto r$, inducing q over the $\binom{100}{3}$ compositions; q is estimated from 10^6 draws and the plan fitted once per instance by (11), which on every instance with $\nu = 0$ reaches a row error below $3 \cdot 10^{-8}$, certifying the remaining inequalities of (5). All rules consume the same $H=5$ local updates from the same draw A^t ; the fixed multiplier is $(m/K)p_i$, the rescaling that presumes uniform observation, and full is the reference $\Phi(p)$. $\arg \min F_w$ is closed-form for least squares and Newton to machine precision for the logistic loss.

C. Adult severity sweep. The five Adult skews omitted from Table 2, same conventions.

β	ν	Δ_{floor}	gain	t
0	.60	.0110	+.0210	44.7
0.25	.60	.0144	+.0181	69.4
0.5	.40	.0129	+.0134	35.2
0.75	0	.0049	+.0050	35.8
1	0	.0003	+.0002	7.1

D. Adult per-race cross-entropy, prevalence regime. Tail-500 over 5 seeds, decaying stepsize.

	White	Black	Asian-Pac.	Amer-Ind.	Other
GAVOT	.339	.179	.368	.175	.073
group-blind avg.	.332	.200	.399	.189	.119
full	.379	.177	.328	.123	.031

Transport improves four of five categories and regresses only the majority group, the expected effect of reweighting toward a category-uniform target; the residual loss on the two smallest categories is the unreachable mass of Cor. 4.